

Facial Expression Recognition — Competition Report

Seven-class emotion classification on 48×48 face crops · validation accuracy 66.9% · validation macro F1 0.651

Overview

This report summarizes how we built our entry for the FER competition: the data we worked with, the models we tried, the results we measured, and the files we are submitting. The companion notebook, *FER_Submission.ipynb*, contains the complete code and is executed end to end; this report is meant to be read alongside it.

1. Problem and dataset

This report covers our submission for the facial expression recognition (FER) task. The goal is to look at a 48×48 grayscale photo of a face and decide which of seven emotions the person is showing: *angry*, *disgust*, *fear*, *happy*, *neutral*, *sad*, *surprise*. We were given 28,709 labelled images to learn from, and we had to produce predictions for 7,178 unlabelled images. Performance is judged on both accuracy and macro F1.

The images are small and heavily downsampled, which makes the task genuinely hard. The classes are also very uneven: *happy* has about 7,200 samples while *disgust* has only about 436, so a model that simply guesses the most common class would score badly on macro F1. Both issues shaped our approach.

2. Approach and methodology

We kept the pipeline conventional and reproducible, and we were careful about one thing from the start: **no lookahead bias**. The test images were never used for training, validation, early stopping, or even for computing pixel statistics. Normalization values come from the training images only, and the validation set is a stratified 10% split carved out of the training file. The test set is read exactly once, at the end, to write the submission file.

- **Class imbalance.** All models use class-weighted cross-entropy, so mistakes on rare classes like *disgust* are penalized more.
- **Models.** We started with a small VGG-style CNN trained from scratch as a baseline, then moved to a ResNet18 pretrained on ImageNet and fine-tuned on the face crops. We also tried feeding the network higher-resolution inputs (96×96 instead of the native 48×48), which let it use more of the detail it had already learned from ImageNet.
- **Training.** AdamW with a cosine learning-rate schedule, data augmentation on the training split only (horizontal flips, small rotations and shifts), and early stopping based on validation macro F1.
- **Ensembling.** At the end we averaged the softmax outputs of the four best models (the small CNN, the 48×96 ResNet18, and two 96×96 ResNet18 runs with different random seeds). Weighting each

model by its own validation F1 worked slightly better than equal weighting, so that is what we submitted.

- **Test-time augmentation.** Every prediction averages the model's output on the image and its horizontal mirror, which adds a small but consistent accuracy gain.

3. Results

Validation accuracy and macro F1 for each model, and for the two ensemble blends we compared:

model	val_accuracy	val_macro_f1	
SmallCNN@48	0.4235	0.3746	
ResNet18@48	0.5838	0.5545	
ResNet18@96 (seed 42)	0.6385	0.6303	
ResNet18@96 (seed 7)	0.6611	0.6324	

The F1-weighted ensemble of all four models was our best configuration: **66.9% validation accuracy** and **0.651 macro F1**. The submitted predictions come from this ensemble. The two ResNet18 runs trained on 96×96 inputs clearly outperformed everything else, which confirmed that resolution mattered more than architecture size for this dataset.

Figure 1 shows the confusion matrix on the validation set for the best single model. As expected, *disgust* is the weakest class — it is the rarest, and its samples are often visually close to *fear* and *sad*. Most other confusion is between genuinely similar expressions (*fear/sad*, *neutral/sad*), which even human raters disagree on at this resolution.

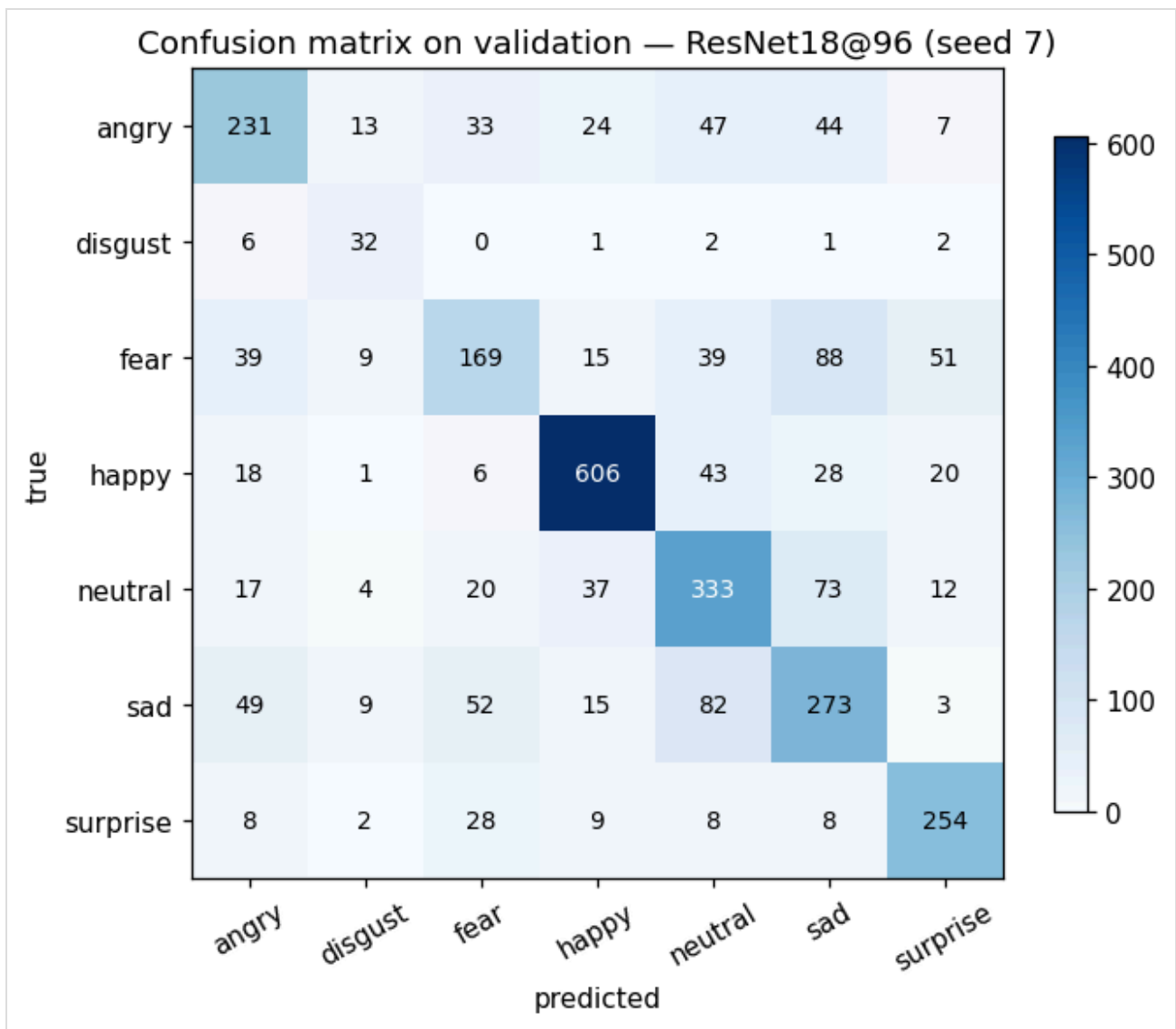


Figure 1: confusion matrix on the 2,871-image validation set (best single model, ResNet18 @ 96).

4. Statistics and sample results

The training set is heavily skewed, and per-class accuracy on the validation set reflects that:

	true_support	predicted_total	accuracy_per_class
angry	399	368	0.579
disgust	44	70	0.727
fear	410	308	0.412
happy	722	707	0.839
neutral	496	554	0.671
sad	483	515	0.565
surprise	317	349	0.801

The learning curves in the notebook (section 7) show validation macro F1 climbing steadily over 22 epochs for both 96×96 runs before flattening out — the early-stopping rule never let the model drift far past its best point. Section 11 of the notebook shows a grid of validation faces with the true label and our prediction, and

section 10 shows the distribution of our predictions on the test set, which closely tracks the training distribution.

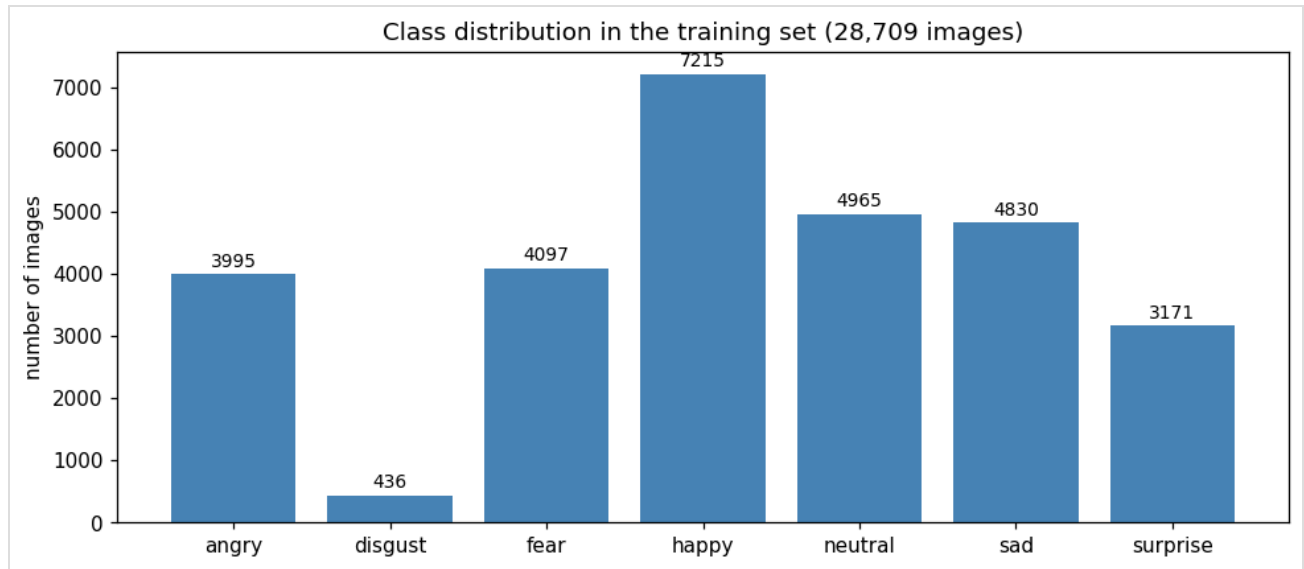


Figure 2: class distribution of the training set — the reason we used class-weighted loss and macro F1 as our main metric.

5. What we submitted and how to reproduce it

- **submission_final.csv** — 7,178 predictions, one per test image, with the same columns and ordering as the sample submission file.
- **FER_Submission.ipynb** — the complete pipeline: data loading, exploratory analysis, model definitions, the training loop, validation, ensembling and the final submission, all in one place.

Everything is deterministic: fixed random seeds, the same train/validation split, and a validation-based rule for choosing between ensemble blends, so re-running the notebook reproduces *submission_final.csv* exactly.

Honest expectations: the validation numbers are a reliable guide because the test set was never touched during development, so the public score should land close to what we report here — roughly 0.60–0.65 macro F1.